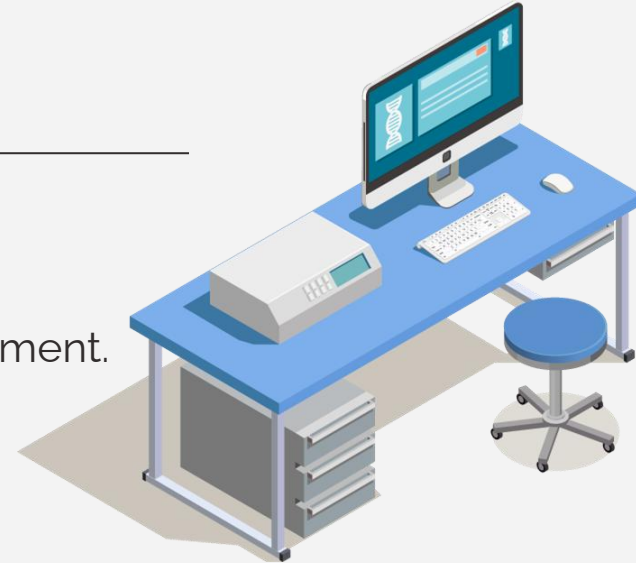




Biomedical Data Acquisition Course (**BMD351**)



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Associate Professor,
Biomedical Informatics Department.



02

Biomedical Signals



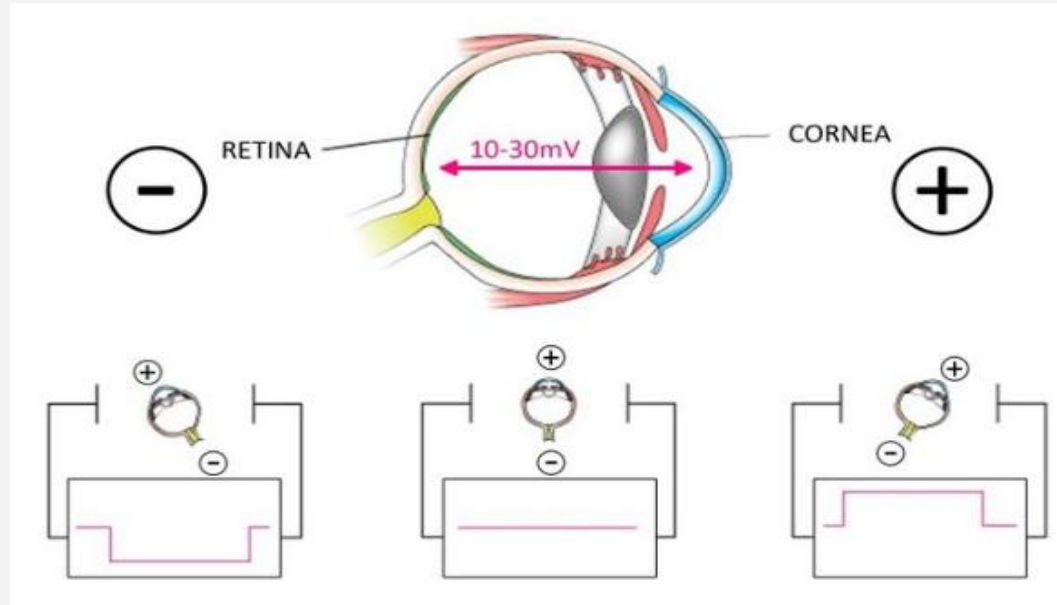
EOG

Eye Signals

Bio-Electric Signals

- Electrical potential generated by the cell membrane potential called action potential and mainly generated by nerve and cell muscle.
- Examples:
 - (1) Electroencephalography (EEG): brain signal recordings.
 - (2) Electromyography (EMG): muscle signal recordings.
 - (3) Electrooculography (EOG): eye signal recordings.
 - (4) Electrocardiography (ECG): heart signal recordings.

Graphical illustration of the corneo-retinal standing potential of the eyeball.



Electrooculography

- **Electrooculography (EOG)** is a technique for measuring the corneo-retinal standing potential that exists between the front and the back of the human eye.
- The resulting signal is called the electrooculogram and the instrument is called electrooculograph.
- Metal superficial disks called electrodes are placed on the top and bottom of the eyes and the right and left.

EOG electrode placement

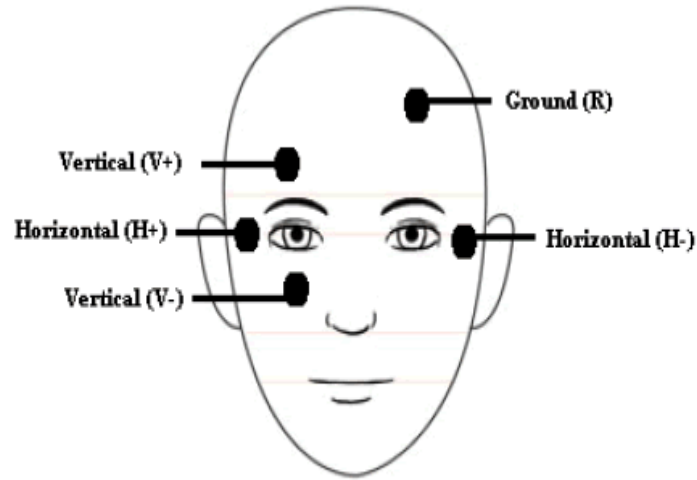
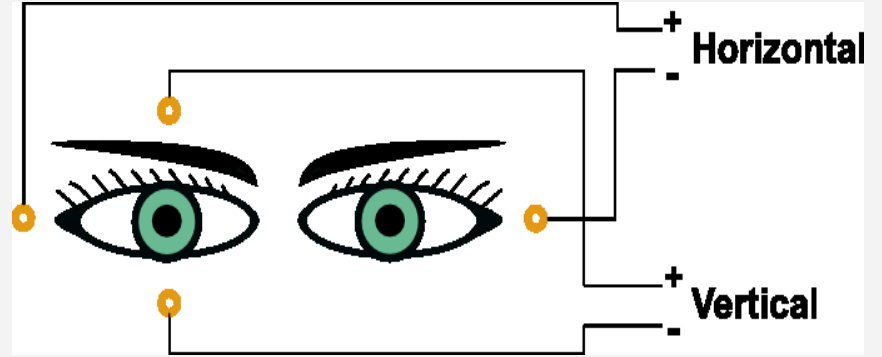
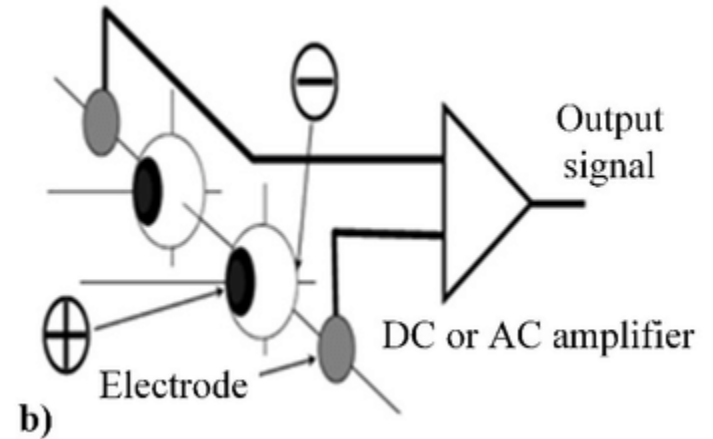
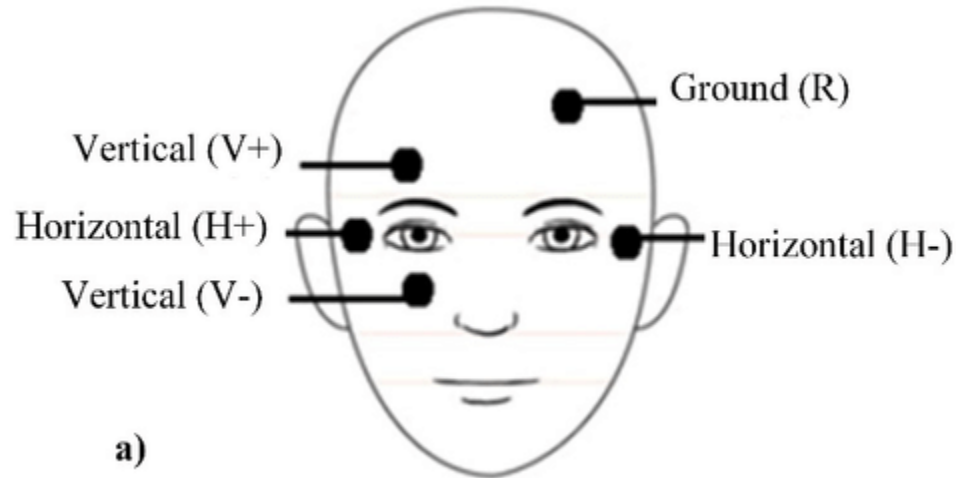


Figure 1. Electrode placement.



EOG device



EOG electrode placement

- **HEOG signals:** obtained from horizontal eye movements .
 - Eye movements from center to left , +ve voltage spike and from center to right -ve voltage spike.
 - **VEOG signals** obtained from vertical eye movements .
 - Eye movements from center to top +ve voltage spike and from center to bottom , -ve voltage spike.
 - Blink Signals obtained due to blinking of eyes.
 - EOG is an Inexpensive, real-time, and reliable test.
 - EOG can be used with with glasses and contact lens
 - There is no obstacles in front of eyes in the EOG test.
-

EOG

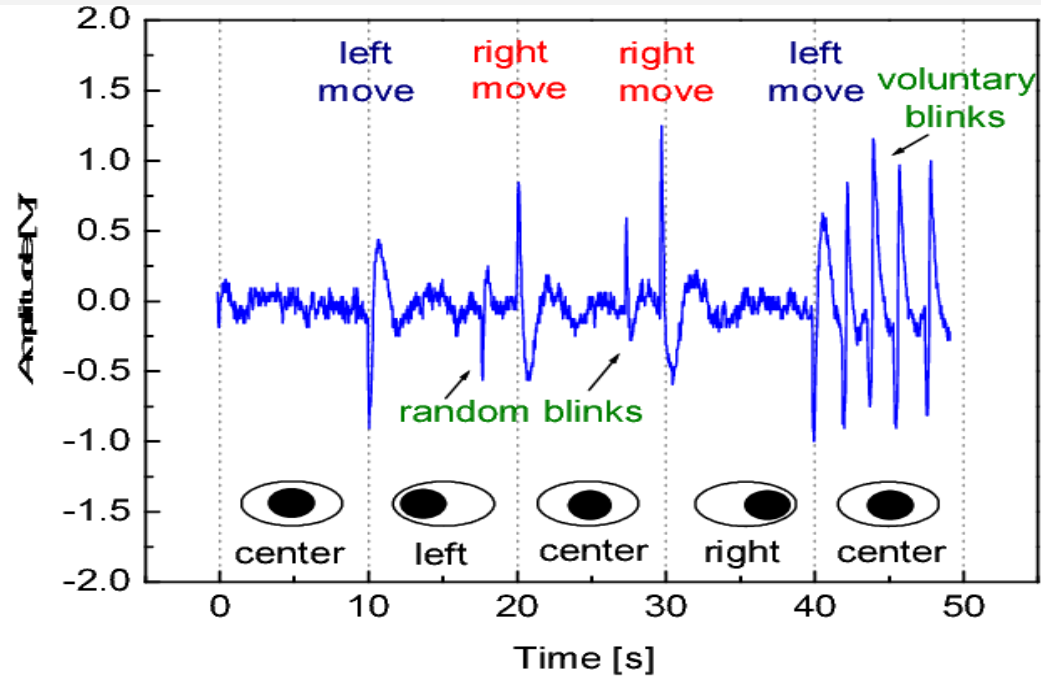
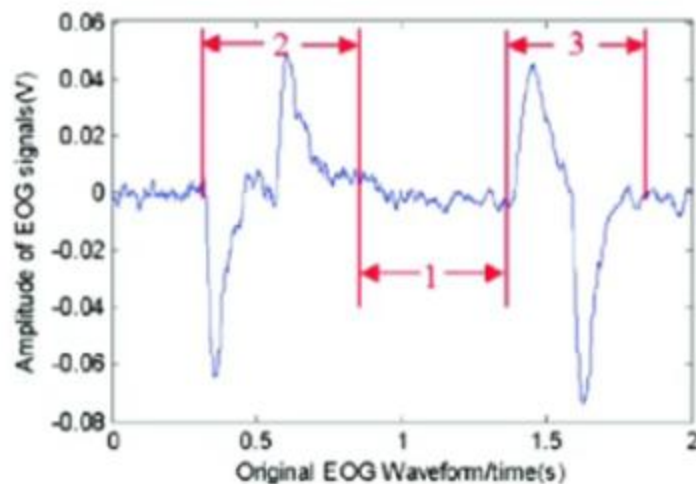


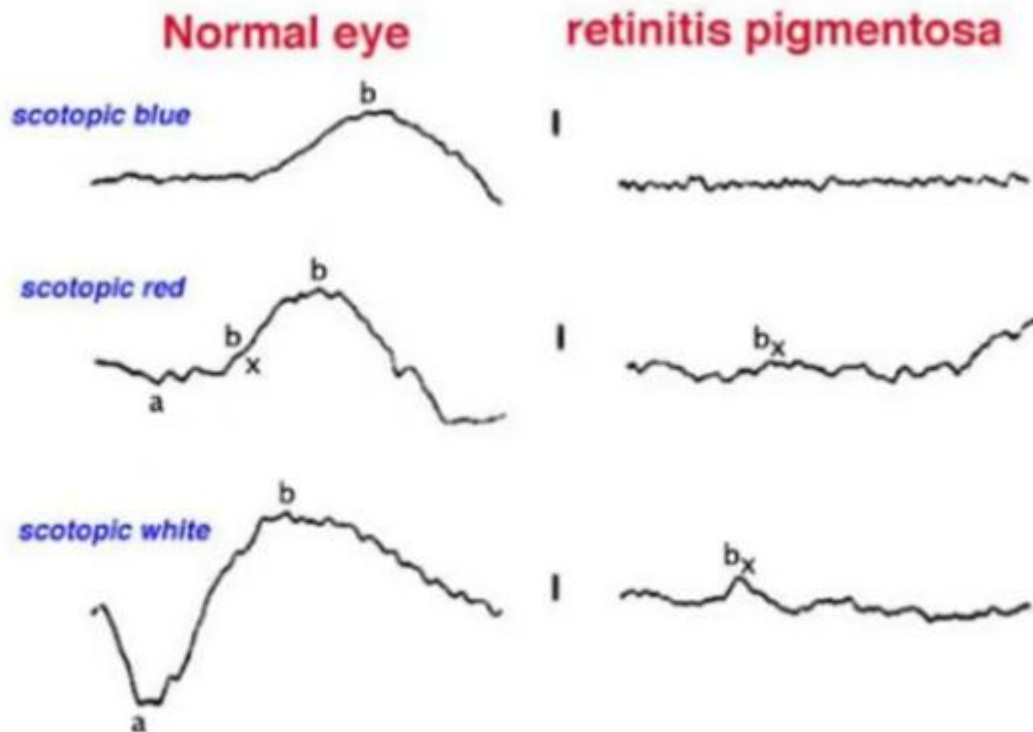
Fig. 5. EOG signals acquired using graphene textile electrodes while eyes are

EOG WAVEFORM

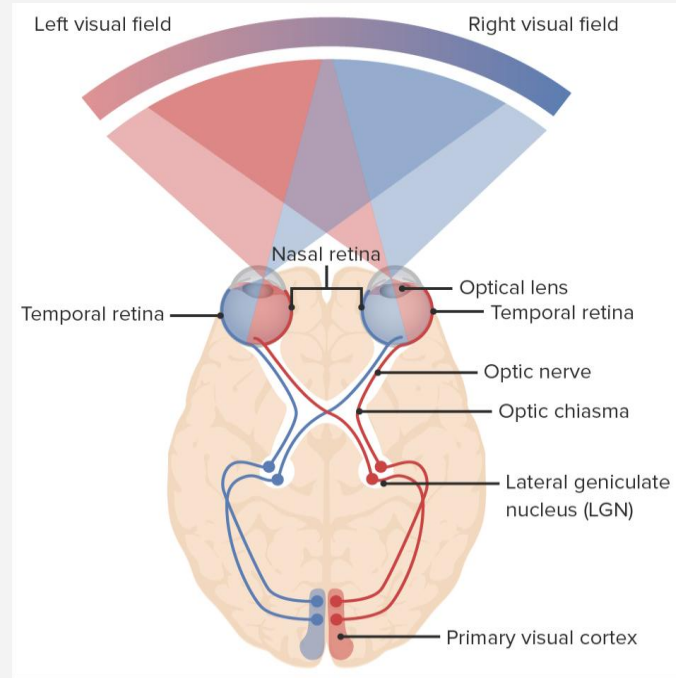
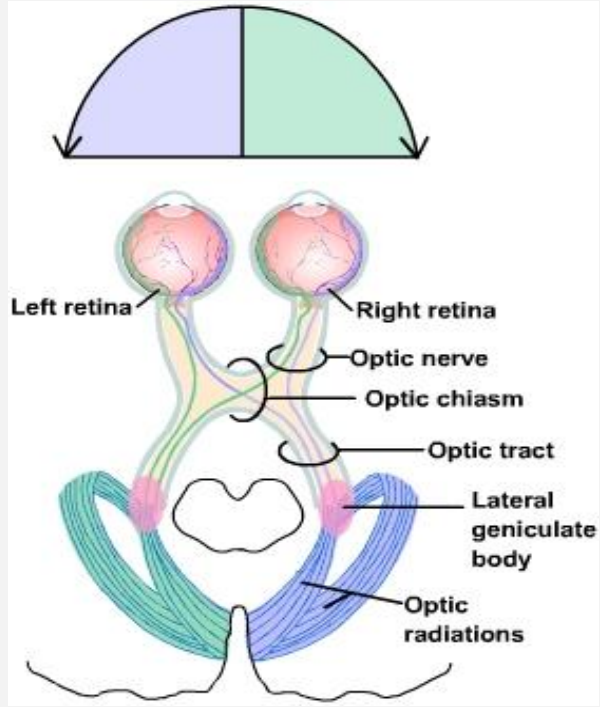
(1) looking straight ahead (2) rolling eyes upward (3) rolling eyes downward



EOG WAVEFORM (NORMAL AND UPNORMAL)



Visual Pathway



Visual Evoked Potential/ Response

- Visual Evoked Potential/Response (VEP/VER) measures the electrical signal generated at the visual cortex in response to visual stimulation.
- VEP depends on the integrity of the visual pathway including eye, optic nerve, optic chiasm, optic tract, optic radiation and cerebral cortex.
- **Visual Evoked Potentials (VEP) Testing:** measures the signals from your visual pathway. Small gold cups called electrodes are pasted to your head to allow us to record those signals..
- **Electrodes:** As per international 10/20 system of recording electroencephalogram. z denotes midline structures:
 - Occipital lobe (Oz, active/positive)
 - Forehead (Fz, reference/negative)
 - Earlobe/vertex (ground/neutral)

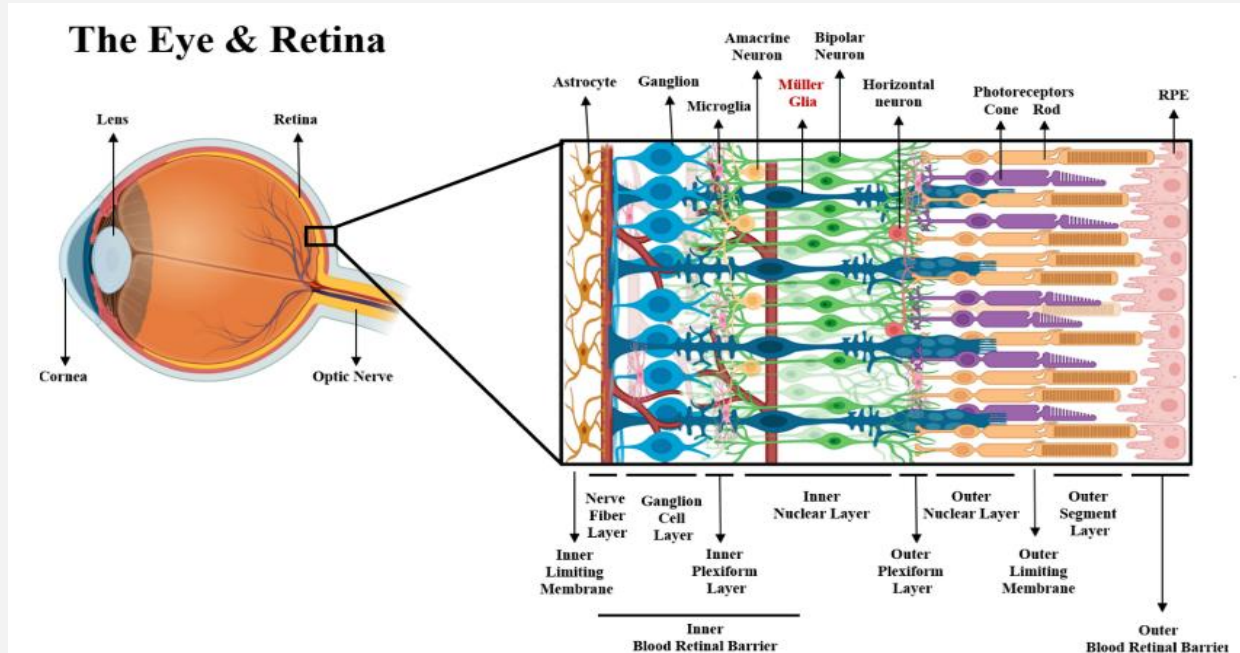
VEP

- The two parts of visual evoked potentials (VEP) testing are **Pattern VEP** and **Flash VEP**:
 - For the Pattern VEP, you will be shown black and white squares that resemble a checkerboard on a computer screen. You will see the squares move and change in size throughout the test. As you're watching this, the signals from your brain will be recorded through the electrodes.
 - For the Flash VEP, we use a machine with a light inside it. You will sit close to the machine and watch as the light inside flashes. This part usually moves along faster than the first part.

Electroretinogram (ERG)

- The electroretinogram (ERG) is a diagnostic test that measures the electrical activity of the retina in response to a light stimulus.
 - The ERG arises from currents generated directly by retinal neurons in combination with contributions from retinal glia.
 - ERGs are often recorded using a thin fiber electrode that is placed in contact with the cornea or an electrode that is embedded within a corneal contact lens. These electrodes permit the electrical activity generated by the retina to be recorded at the corneal surface.
-

ERG



EOG sensors



Price: €120,00

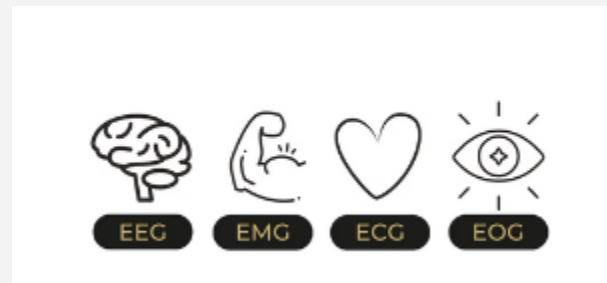
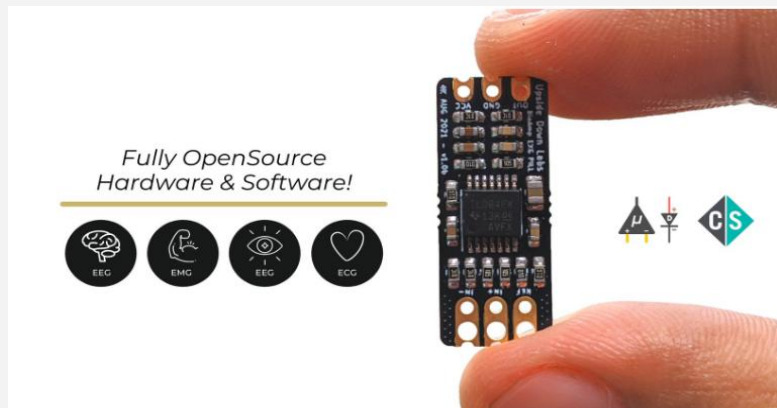
✓ The biosignalsplux Electrooculography (EOG) sensor has been specially designed for seamless EOG data acquisition. Either used by itself or in combination with an eye tracker, our sensor can provide additional insight into your subjects' eye gaze patterns.

✓ The bipolar configuration, with two measurement electrodes, detects the electrical potentials in the specific temporal or facial region of choice, with respect to a reference electrode (placed in an area of low bioelectrical activity). The resulting signal is the amplified difference between these two leads, eliminating the common unwanted signals. Its convenient form factor enables a discrete application in the typical EOG electrode placement locations.

Biomedical Data Acquisition Sensor

BioAmp EXG Pill

Professional-grade analog front-end amplification for ECG, EMG, EOG, and EEG biosensing on one tiny board



Projects

- **Electrooculography (EOG) Speller (Assistive Keyboard, Mouse, etc.)**
- **Assistive people with disabilities using eye movements and blinks.**
- **Eye diseases diagnosis based on eye signals (التهاب الشبكية الصباغي).**

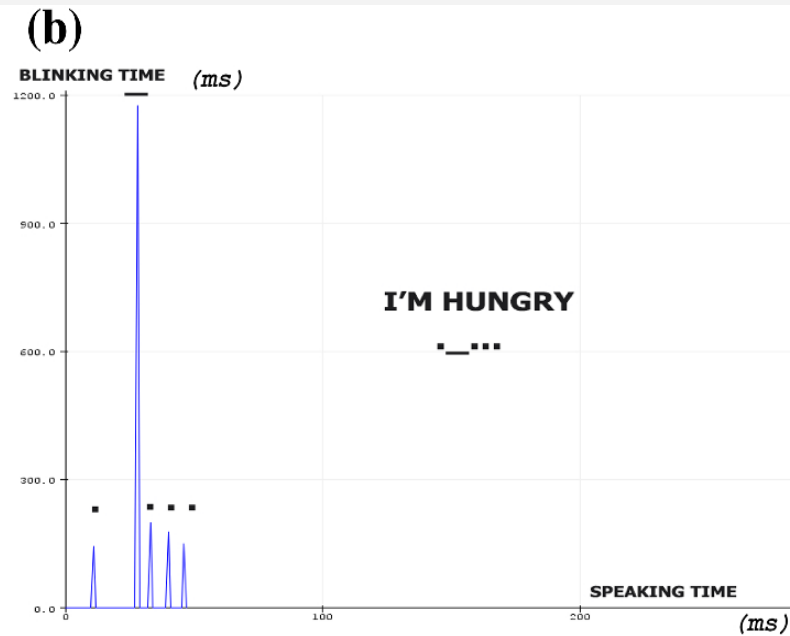
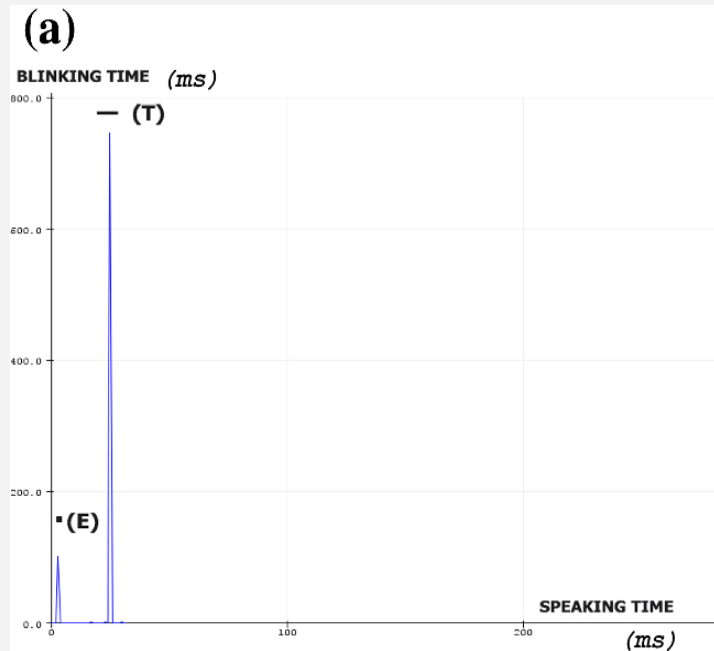
Regular Paper | [Published: 14 June 2021](#)

Morse glasses: an IoT communication system based on Morse code for users with speech impairments

[Nayera Tarek](#), [Mariam Abo Mandour](#), [Nada El-Madah](#), [Reem Ali](#), [Sara Yahia](#), [Bassant Mohamed](#), [Dina Mostafa](#) & [Sara El-Metwally](#) 

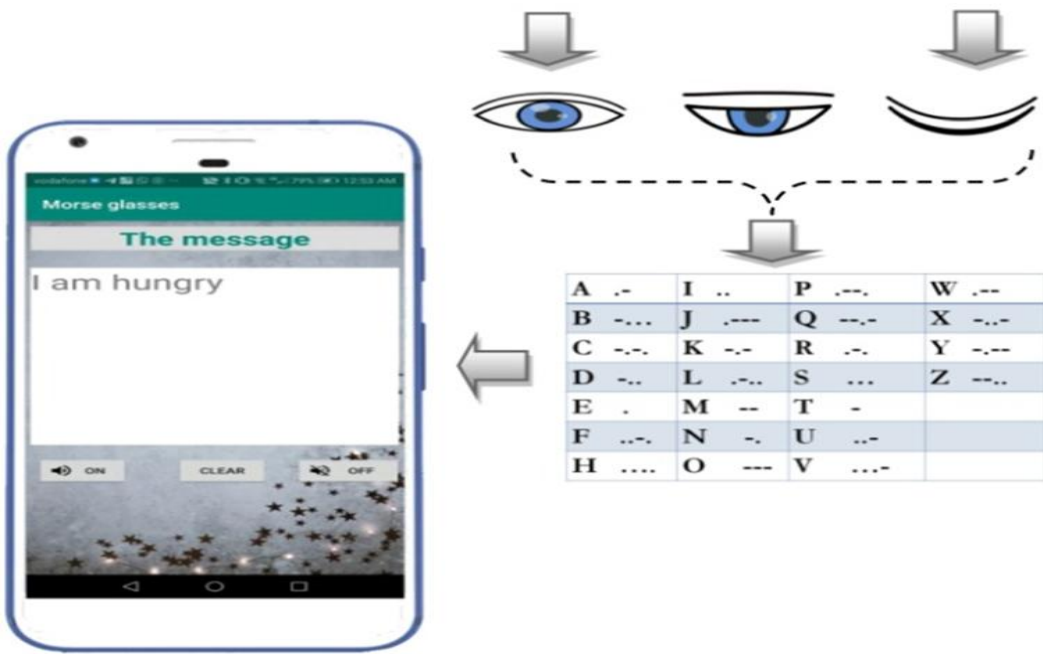
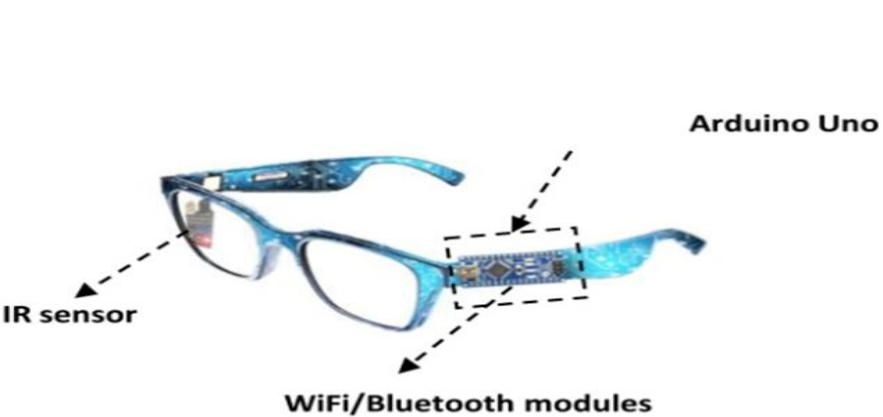
[Computing](#) **104**, 789–808 (2022) | [Cite this article](#)

Note



Eye's blink < 0.6 sec. = Dot = . 		Eye's blink > 0.6 sec. = Dash = - 	
A	 	N	 
B	   	O	   
C	   	P	   
D	  	Q	   
E		R	  
F	   	S	  
H	   	T	
I	 	U	  
J	   	V	   
K	  	W	  
L	   	X	   
M	 	Y	   
Z	   	.	    
Space		     	

Eye's blink < 0.6 sec. = Dot = . 	Eye's blink > 0.6 sec. = Dash = - 
I feel pain, I want to take my medicine    	
I am hungry    	
I am thirsty    	
I need help    	
I feel hot    	





Morse glasses

The message

I am hungry

A ● -	J ● - - -	S ● ● ●
B - ● ● ●	K - ● -	T -
C - ● - ●	L ● - ● ●	U ● ● -
D - ● ●	M - -	V ● ● ● -
E ●	N - ●	W ● - -
F ● ● - ●	O - - -	X - ● ● -
G - - ●	P ● - - ●	Y - ● - -
H ● ● ● ●	Q - - ● -	Z - - ● ●
I ● ●	R ● - ●	





Thanks!
